

TOMSK, Russian Federation

WMO: 294300

Lat: 56.500N

Lon: 84.917E

Elev: 139

StdP: 99.66

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -36.4	(c) -33.3	(d) -39.2	(e) 0.1	(f) -36.2	(g) -36.2	(h) 0.1	(i) -33.2	(j) 6.7	(k) -12.0	(l) 5.9	(m) -12.9	(n) 0.5	(o) 230	(p) 0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 9.7	(c) 28.7	(d) 19.6	(e) 27.1	(f) 18.7	(g) 25.5	(h) 17.9	(i) 21.0	(j) 26.2	(k) 20.0	(l) 25.2	(m) 19.1	(n) 23.7	(o) 1.4	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.2	23.3	18.2	13.3	22.5	17.2	12.5	21.6	61.7	26.3	58.3	25.2	55.0	23.8	25.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5.1	4.0	3.3	-38.8	31.7	3.5	1.4	-41.3	32.7	-43.4	33.5	-45.4	34.3	-48.0	35.3
			-38.6	22.5	3.6	1.2	-41.2	23.3	-43.3	24.0	-45.3	24.6	-47.9	25.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.3	-18.4	-14.3	-6.2	2.5	10.4	16.7	18.9	15.9	9.5	2.1	-8.0	-14.9
	DBStd	14.14	8.67	7.37	6.24	5.41	5.51	4.41	3.23	3.70	4.47	5.10	7.90	8.42
	HDD10.0	3991	881	681	502	229	64	5	0	3	64	248	541	772
	HDD18.3	6336	1139	914	760	474	251	79	33	92	268	503	791	1031
	CDD10.0	802	0	0	0	6	77	207	274	186	48	4	0	0
	CDD18.3	105	0	0	0	0	6	32	49	17	1	0	0	0
	CDH23.3	899	0	0	0	1	88	285	372	139	15	0	0	0
	CDH26.7	185	0	0	0	0	22	59	76	26	2	0	0	0
Wind	WSAvg	1.4	1.4	1.5	1.5	1.7	1.6	1.2	1.1	1.0	1.1	1.3	1.5	1.5
Precipitation	PrecAvg	527	31	19	20	31	43	61	66	75	46	50	49	38
	PrecMax	739	67	45	45	79	92	132	150	158	143	105	87	76
	PrecMin	378	11	0	2	8	10	17	1	14	12	14	11	12
	PrecStd	77	13	10	10	17	19	28	33	39	28	20	18	15
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.2	1.3	8.9	20.7	29.4	30.5	30.7	29.7	25.5	17.7	7.3	1.0
		MCWB	-0.8	-0.4	4.1	11.6	17.7	19.0	21.2	20.7	17.3	10.7	5.3	-0.2
	2%	DB	-2.4	-0.5	5.9	16.9	25.7	28.3	28.7	26.8	21.7	14.4	4.8	-0.4
		MCWB	-3.2	-1.8	2.0	9.7	15.2	18.7	20.4	19.2	14.9	9.4	3.3	-1.0
	5%	DB	-4.2	-2.3	3.9	13.4	22.9	26.6	27.0	24.6	18.9	11.7	2.9	-2.0
		MCWB	-4.9	-3.4	0.8	7.7	13.6	18.1	19.5	18.0	13.6	8.4	1.8	-2.5
	10%	DB	-6.2	-4.2	2.1	10.6	19.9	24.8	25.4	22.6	16.3	9.0	1.2	-3.9
		MCWB	-6.9	-5.0	0.0	6.2	12.4	17.3	18.6	17.0	12.3	6.5	0.3	-4.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.8	0.2	4.5	12.3	18.7	21.3	22.7	21.5	17.8	12.0	6.0	0.2
		MCDB	-0.1	1.0	8.3	19.3	28.2	27.3	27.9	27.5	23.5	15.0	7.0	0.7
	2%	WB	-3.0	-1.7	2.4	10.2	16.3	19.9	21.5	20.1	15.6	10.2	3.5	-1.0
		MCDB	-2.4	-0.7	5.0	16.0	23.6	25.7	26.6	25.2	20.6	13.3	4.4	-0.6
	5%	WB	-4.9	-3.3	1.2	8.1	14.5	18.9	20.5	18.9	14.1	8.5	1.8	-2.5
		MCDB	-4.2	-2.4	3.3	12.7	21.1	24.7	25.3	23.2	17.9	11.6	2.8	-2.0
	10%	WB	-6.8	-5.0	0.1	6.3	13.0	17.9	19.6	17.7	12.8	6.7	0.4	-4.5
		MCDB	-6.2	-4.2	2.0	10.3	18.9	23.4	24.2	21.6	15.9	8.9	1.1	-3.9
Mean Daily Temperature Range	5% DB	MDBR	7.1	8.6	9.8	9.5	10.9	10.2	9.7	9.4	8.8	6.7	6.0	6.7
		MCDBR	6.9	7.5	10.7	13.6	15.2	13.3	12.1	12.7	13.2	11.3	5.9	6.9
		MCWBR	6.9	6.5	7.5	8.0	7.4	5.7	5.2	5.9	6.9	6.9	4.8	6.8
	5% WB	MCDBR	7.1	7.0	9.0	13.0	13.8	11.7	10.7	11.2	11.9	10.5	5.8	6.9
		MCWBR	7.1	6.4	6.6	8.0	7.6	5.6	5.3	5.8	6.6	6.7	4.9	6.9
Clear-Sky Solar Irradiance	taub	0.292	0.325	0.374	0.401	0.407	0.406	0.436	0.408	0.371	0.351	0.319	0.293	
	taud	1.994	1.966	1.911	2.062	2.218	2.304	2.216	2.285	2.335	2.238	2.050	1.950	
	Ebn at Noon	547	688	750	805	831	837	798	797	774	671	507	422	
	Edh at Noon	66	102	141	141	129	120	129	113	93	79	62	53	
All-Sky Solar Radiation	RadAvg	0.78	1.77	3.17	4.39	4.99	5.78	5.26	4.23	2.78	1.38	0.65	0.46	
	RadStd	0.08	0.12	0.22	0.34	0.50	0.58	0.35	0.34	0.39	0.11	0.11	0.09	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page